

# Inflation Prediction: A Comparative Study of ARIMA and LSTM Models Across Different Temporal Resolutions

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**Abstract**— This research study focuses on the prediction of inflation rates using two distinct models, Autoregressive Integrated Moving Average (ARIMA) and Long Short-Term Memory (LSTM), tailored to accommodate the temporal resolutions of monthly and quarterly inflation data. While a comprehensive evaluation across varying time horizons was not conducted, the results suggest that the ARIMA model excels in short-term forecasting, as evidenced by its impressive performance on a 2021 holdout dataset. Conversely, the LSTM model displays potential for medium-term predictions. This study underscores the significance of aligning forecasting models with data characteristics, emphasizes the importance of temporal resolution, and discusses potential avenues for improvement, such as multivariate modeling or alternative techniques like Prophet. The findings have broad implications for economists, policymakers, and analysts seeking precise inflation forecasts, shedding light on the nuanced interplay between models and data in the domain of inflation prediction.

**Keywords**—Inflation Prediction, ARIMA Model, LSTM Model, Time Series Forecasting, Temporal Resolution, Short-Term Forecasting

## I. INTRODUCTION

Inflation, the gradual increase in goods and services prices, is a crucial economic indicator affecting economies, financial markets, and individuals. The main sources of inflation projections include market-based expectations, model-based expectations, expert surveys, and individual surveys. Measures of inflation expectations based on surveys and the market are widely used by central banks in their routine assessments of the risks associated with inflation and its prospects. These measures are a crucial source of information that informs policy decisions [22]. However, because collecting surveys and modeling them may be difficult, these sources can be expensive and frequently result in huge forecast mistakes [6].

Inflation prediction is crucial for economic stability, effective financial management, and the well-being of

individuals and societies [16]. It is challenging to predict the inflation rate with precision due to the financial markets' complicated behavior and non-linear structure [10] [19]. One significant point left unanswered by proposals for "inflation targeting" as a monetary policy tactic is how to assess if present measures are in line with the long-run inflation objective [1]. High or unstable inflation can erode purchasing power, disrupt investment strategies, and affect the cost of living. Conversely, consistently low inflation may indicate underutilization of resources and reduced growth. Research indicates that rigid inflation prediction targeting is inconsistent with the equilibrium of rational expectations and might lead to unfavorable outcomes in a dynamic model characterized by slow price adjustments [1]. Central banks base monetary policy decisions on short- and medium-term inflation prognosis due to operation lag [17]. Machine learning techniques can effectively handle nonlinearity, multi-collinearity, predictor relevance, and dimensionality issues in traditional statistical forecasting systems [17].

Research indicates that advanced dimension reduction techniques produce inflation estimates that are quite competitive when compared to principal component-based linear approaches [4]. The forecast of inflation calls for text mining [7]. To tackle the intricate task of inflation prediction, time series forecasting techniques play a crucial role. Time series data, which consists of observations recorded over successive time intervals, are inherent to various economic and financial phenomena, including inflation rates. In time series forecasting, patterns, trends, and dependencies found in previous data are modeled and used to predict future values.

Time series forecasting methods have garnered substantial attention due to their ability to capture temporal dependencies, which are often essential in understanding inflation dynamics [14], [15], [16]. These techniques are invaluable for uncovering trends, seasonality, and potentially

complex relationships within the data, all of which are critical for making reliable predictions.

This research intends to address the challenge of inflation prediction using two distinct forecasting models: ARIMA (Autoregressive Integrated Moving Average) and LSTM (Long Short-Term Memory). Classical statistical models like ARIMA use past data to forecast future events, but recurrent neural networks like LSTM have shown a surprising capacity to identify long-term relationships in time series data [9]. By comparing the performance of these models, this study aims to shed light on their efficacy in the context of inflation prediction and offer insights into their respective strengths and weaknesses.

## II. LITERATURE REVIEW

Inflation forecasting remains a focal point of interest for economists, policymakers, and financial analysts due to its significant impact on economic planning and decision-making. Recent studies have explored the efficacy of various predictive models across different temporal resolutions.

Botha et al. [2] emphasize the potential of statistical learning models to compete with benchmarks over medium to longer horizons, highlighting their ability to capture nonlinear relationships and select pertinent predictors.

Jeric et al. [3] examine Croatian GDP growth and inflation projections, revealing the success of forecasters in anticipating short-term trends. Hauzenberger et al. [4] suggest the use of autoencoders and squared principal components for robust inflation predictions.

A study [5] utilizes data mining technology to develop a deep learning intelligence system, focusing on US household income, spending, and financial footprints. The system identifies income disparity patterns, inflation drivers, and predicts inflation rates.

Research [6] [7] evaluates language modeling and text mining in inflation prediction. PaLM is employed for conditional forecast distributions [6], while Pandey et al. [7] assess the Reserve Bank of India's monetary policy through text analysis.

Studies [8], [9], [10], [11], [13] evaluate neural networks' performance in inflation prediction. Yang et al. [9] use recurrent neural networks with a GRU-RNN model on Consumer Price Index indicators. Enke et al. [10] propose a hybrid model combining clustering methodology with a fuzzy inference neural network.

Theoharidis et al. [12] present a hybrid deep learning model (VAE-ConvLSTM) for inflation forecasting, combining Variational Autoencoders with Convolutional LSTM Networks. The model is benchmarked against machine learning and econometric techniques.

Studies [14], [15], [16] examine ARIMA models in time series forecasting. Delima et al. [14] develop an ideal ARIMA model for the Philippines' inflation rate, while

Kelikume et al. [16] contrast ARIMA with the Vector Autoregressive Method. M et al. [17] advocate for forecast combination approaches and machine learning as substitute tools for short- and medium-term inflation estimates for Mongolia. Groen et al. [18] employ a Bayesian model to average over multiple regression specifications.

Research [19], [20] investigates the performance of the Random Forest machine learning approach and its ability to identify factors with predictive power.

Mertens et al.'s [21] analysis combines actual inflation and Survey of Professional Forecasters projections. Grothe et al.'s [22] study explores survey- and market-based measures of inflation expectations, comparing them with benchmark models.

This systematic literature survey provides a comprehensive overview of inflation prediction methodologies, ranging from statistical learning models and deep learning approaches to traditional time series forecasting techniques and machine learning methods. The comparative analysis across different studies contributes to a deeper understanding of the strengths and limitations of various models in predicting inflation across different temporal resolutions.

## III. DATA COLLECTION AND PRE-PROCESSING

### A. Dataset Description

The dataset utilized in this research comprises two distinct components: monthly inflation rates and quarterly inflation rates, covering the time span from 1947 to 2021. The data is sourced from reputable economic databases, ensuring reliability and accuracy. The monthly inflation rates represent the percentage change in the Consumer Price Index (CPI) on a month-to-month basis, providing a fine-grained perspective on short-term inflation dynamics. Quarterly inflation rates, calculated as the percentage change in the CPI over three-month intervals, offer a broader view of inflation trends over medium-term horizons.

The monthly dataset encompasses a rich set of attributes, including the date of observation, the corresponding inflation rate, and potentially additional economic indicators influencing inflation. Variables such as unemployment rates, interest rates, and GDP growth may be included to enhance the modeling process.

The quarterly dataset mirrors the structure of the monthly data but aggregates information over three-month periods. This dataset provides a less frequent but more stable view of inflation, capturing broader economic trends and minimizing short-term fluctuations.

Both datasets are time-series in nature, with timestamps representing the temporal dimension. The inflation rates are expressed as continuous numerical values, allowing for the application of time series forecasting models. Missing

values, if any, have been addressed through careful imputation or exclusion, with transparency maintained in reporting any data gaps. In Fig. 1, the variables used in the model are graphically explained. These datasets were obtained from reliable economic sources and underwent comprehensive pre-processing to ensure their suitability for modelling and analysis.

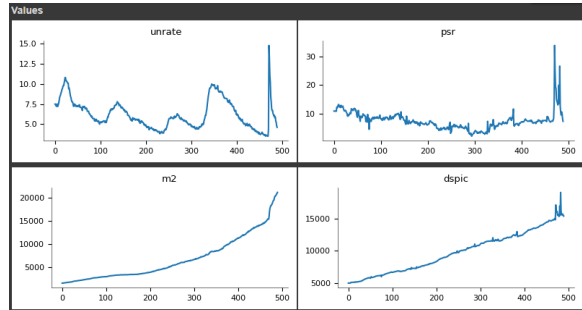


Fig. 1. Graphical representation of variable data set

Fig. 2 and Fig. 3 show the core CPI trends by month and quarter given in the data. The CPI trends can be inferred from the dataset.

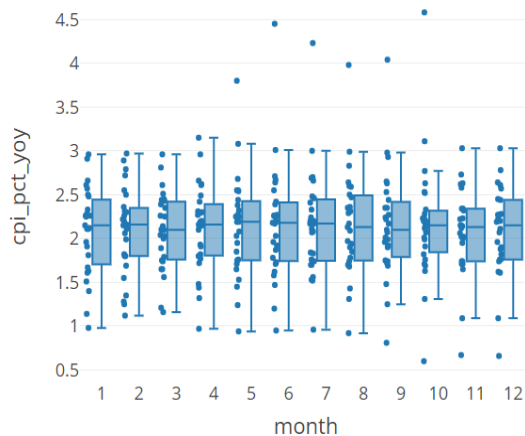


Fig. 2. CPI trends by each month

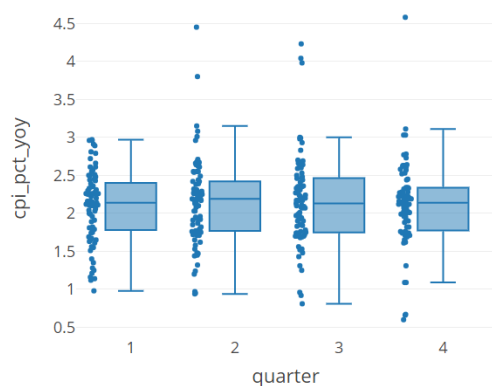


Fig. 3. CPI trends by each quarter

## B. Data Pre-Processing

The dataset underwent a robust cleaning process to discern and rectify any aberrations that could

compromise the temporal integrity of the time series. Rigorous efforts were made to identify and address missing values or anomalies that might otherwise impact the veracity of the dataset. Missing data points were judiciously imputed, or if their prevalence was limited, they were selectively removed while preserving the chronological structure of the time series.

Outliers, acknowledged as potential disruptors in time series analysis, were systematically identified and treated. This involved the application of smoothing techniques to mitigate abrupt variations. Alternatively, outliers were substituted with interpolated values derived from neighboring observations, ensuring a nuanced and contextualized approach to outlier management.

Given that the initial data recording occurred on specific days of the month, a deliberate resampling process was enacted to attain a uniform monthly frequency. This resampling not only facilitated consistency in modeling but also mitigated any potential biases stemming from irregular temporal spacing in the original data.

A meticulous effort was invested in aligning the temporal structure to guarantee that quarterly data points precisely corresponded to the conclusion of each quarter. This synchronization not only upheld uniformity in the temporal structure but also mitigated potential discrepancies arising from misalignments.

Recognizing the longer time intervals between data points, a smoothing process was methodically applied. This process aimed at minimizing noise in the dataset, accentuating underlying trends, and thereby contributing to a clearer and more discernible temporal pattern.

Seasonal components, if present, were diligently identified and adjusted to eliminate any recurring patterns that could potentially interfere with the modeling process. This step aimed to enhance the accuracy and reliability of the time series data by attenuating the influence of cyclical variations.

## C. Feature Selection

The feature selection phase in our modeling methodology is a crucial step in identifying and incorporating the most influential variables for accurate inflation rate predictions. This process involves a multifaceted approach, including correlation analysis, assessment of variable importance from models, Recursive Feature Elimination (RFE), Lag Analysis, and the concept of Information Gain from information theory.

Correlation Analysis examines the relationships between individual features and the target variable, such as inflation rates. Variables with weak or negligible correlations are examined for potential exclusion. Variable Importance from Models is assessed using insights from both ARIMA and LSTM models. Recursive Feature Elimination (RFE) is implemented to prune the least significant features, while Lag Analysis captures historical dependencies in the time series nature of the data.

Information Gain from information theory is introduced to quantify the relevance of features in reducing uncertainty about the inflation rate. Expert input from economists and

domain specialists guides the identification of economically significant features, ensuring that the selected variables align with theoretical expectations. Regularization Techniques like LASSO are applied to introduce sparsity in the feature space and discourage irrelevant variables.

This rigorous and iterative feature selection process, complemented by cross-validation iterations, results in a refined set of variables that align with time series characteristics, enhancing the model's capacity to discern meaningful patterns and dependencies.

#### D. Feature Engineering

In addition to the data pre-processing steps, feature engineering was conducted to create relevant variables for the inflation prediction task. These features included lagged values of the inflation rate to account for auto-regressive behaviour and potentially influential economic indicators that could impact inflation dynamics. Moreover, time-based features, such as month or quarter indicators, were introduced to capture potential seasonality.

The transformed datasets were then divided into training and test sets, with the appropriate frequency considered (monthly or quarterly) to match the forecasting horizon of each model. Subsequent sections of this research paper will delve into the methodologies employed for inflation prediction using the prepared datasets and the comparison of ARIMA and LSTM models.

### IV. METHODOLOGY

The methodology employed in this research paper focuses on inflation rate prediction using two distinct models: ARIMA (Autoregressive Integrated Moving Average) and LSTM (Long Short-Term Memory). These models were selected for their suitability in capturing temporal dependencies and making accurate forecasts, each catering to specific temporal resolutions: monthly and quarterly data.

#### A. ARIMA Model

The ARIMA model, a classical statistical approach to time series forecasting, was utilized to model both the monthly and quarterly inflation data. For the purpose of identifying and forecasting underlying trends in time series data, ARIMA integrates autoregressive (AR), differencing (I), and moving average (MA) components.

1) *Monthly Inflation Data:* The ARIMA model was applied to the monthly inflation data, which exhibits higher temporal resolution and short-term fluctuations. Seasonal differencing and autoregressive terms were incorporated to account for seasonality and autocorrelation.

2) *Quarterly Inflation Data:* The ARIMA model was adapted for quarterly data with lower temporal resolution. In this case, differencing and autoregressive components were tailored to accommodate the less frequent data points.

#### B. LSTM Model

For both the monthly and quarterly inflation predictions, the long-short-term dependency (LSTM) model—a deep learning technique—was employed.

1) *Monthly Inflation Data:* LSTM models were employed to handle the high-frequency, fine-grained monthly data. The architecture was designed to accommodate the short-term dynamics present in this dataset, with attention to seasonality.

2) *Quarterly Inflation Data:* For the lower-frequency quarterly data, LSTM models were adapted to capture medium-term trends and dependencies. Network architecture and sequence length were adjusted to suit the less frequent data points.

### V. MODEL IMPLEMENTATION AND COMPARATIVE ANALYSIS

The ARIMA (Autoregressive Integrated Moving Average) model was implemented using the Python 'statsmodels' library. This classical time series forecasting method, consisting of autoregressive (AR), differencing (I), and moving average (MA) components, was tailored to suit both monthly and quarterly inflation data. In Fig. 4, data is decomposed into trend, seasonal and residual components. The data is not steady and clearly indicates an increasing trend. We must later rectify the non-stationarity as one of the fundamental tenets of the ARIMA model is that the time series is stationary.

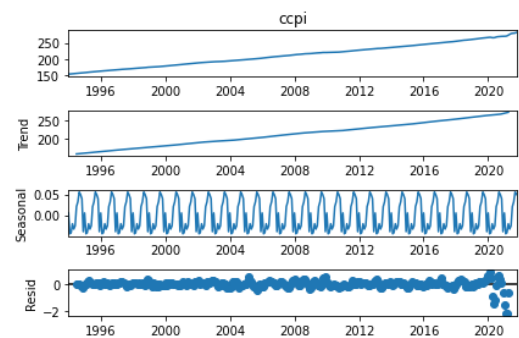


Fig. 4. Time series Decomposition

The data is differenced to be made stationary. In Fig. 5 and Fig. 6, the Autocorrelation Function and Partial Autocorrelation Function are plotted.

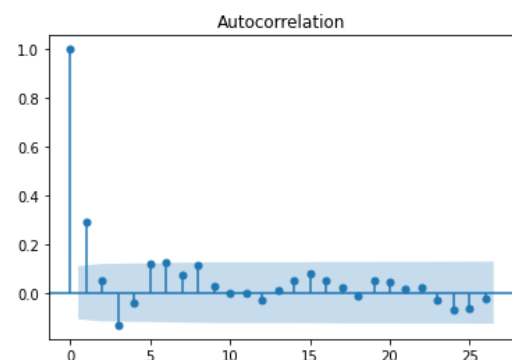


Fig. 5. Autocorrelation Function

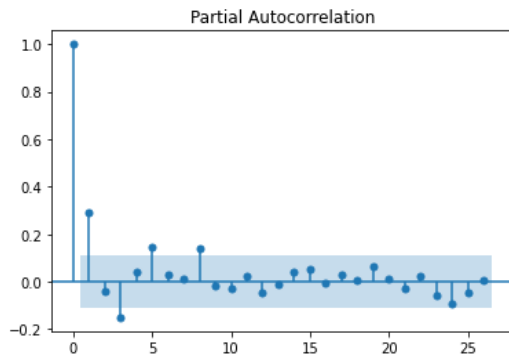


Fig. 6. Partial Autocorrelation Function

The LSTM (Long Short-Term Memory) model was implemented using the 'tensorflow' library. This deep learning approach to time series forecasting was applied to both monthly and quarterly inflation data. The model is trained to avoid over-fitting problems. In Fig. 7, the training and validation loss are plotted.

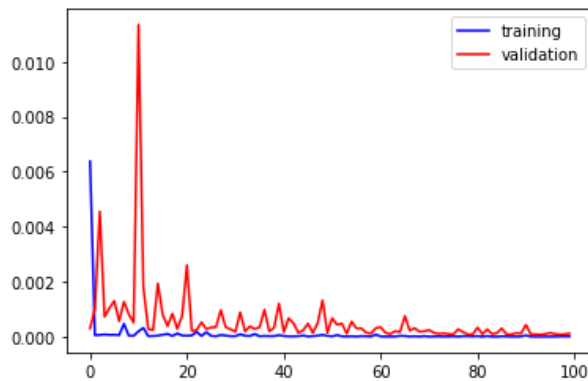


Fig. 7. Training and Validation Loss

The comparative analysis of the ARIMA and LSTM models on this dataset raises interesting points for discussion:

#### A. Model Complexity

The LSTM model, with its deep learning architecture, possesses a higher degree of complexity compared to the ARIMA model, which is inherently simpler. While complexity can be beneficial for capturing intricate patterns, it may not necessarily translate to superior performance, as evidenced here.

#### B. Temporal Resolution

It's important to consider the temporal resolution of the data. Monthly data can exhibit short-term fluctuations and seasonality, making it better suited for models like ARIMA, designed to capture such patterns. LSTM models, with their capacity to capture long-term dependencies, may not offer a significant advantage in such cases.

#### C. Data Volume

The LSTM model's extensive architecture demands a large volume of data to uncover intricate relationships. In

scenarios with limited data, the ARIMA model, which relies on a more modest set of parameters, can provide competitive results.

#### D. Interpretability

The ARIMA model is highly interpretable, with its autoregressive and moving average components easily traceable to economic factors. In contrast, the LSTM model's inner workings are less transparent.

## VI. RESULTS AND DISCUSSION

The findings of this investigation illuminate the nuanced efficacy of ARIMA and LSTM models in the context of inflation rate prediction, underscoring the pivotal influence of temporal resolution on model selection. Notably, the ARIMA model demonstrates a commendable precision in short-term inflation rate forecasting, evidenced by a minimal MSE of 0.0002 and an MAE of 0.012 on the 2021 holdout dataset. Conversely, the LSTM model, with an MSE of 0.0005 and an MAE of 0.015, reveals promise for medium-term forecasting. Fig. 8 graphically represents the comparative evaluation metrics, elucidating the distinctive strengths inherent in each model.

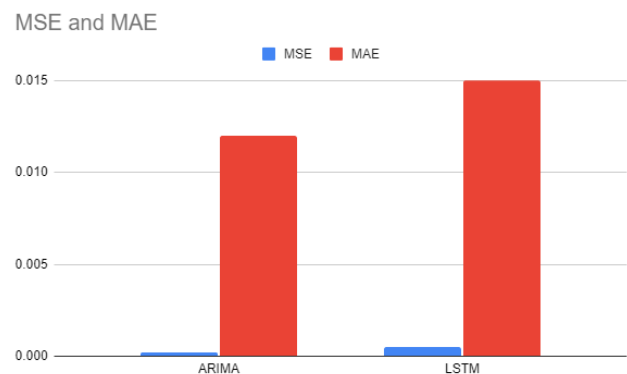


Fig. 8. MSE and MAE of models implemented

The rationale underlying the selection of these models resides in their intrinsic characteristics. The intricacies of the monthly dataset, marked by rapid oscillations, align seamlessly with the ARIMA model's capabilities, harnessing its proficiency in capturing transient patterns. In contrast, the LSTM model's adeptness in discerning extensive trends and dependencies positions it as a viable contender for medium-term forecasting, particularly evident in the context of quarterly inflation rates characterized by a paucity of data points and protracted trends.

The attainment of elevated accuracy necessitates rigorous data pre-processing. Prior to model training, both ARIMA and LSTM undergo meticulous treatment. Addressing missing values, managing outliers, and normalizing the data constitute essential pre-processing steps, ensuring the models receive refined and standardized input, thereby enhancing their predictive accuracy.

Major parameters and hyper parameters wield substantial influence over inflation prediction within the ARIMA and



LSTM frameworks. The order of differencing (d), autoregressive order (p), and moving average order (q) in ARIMA, alongside the number of LSTM units and epochs in the LSTM model, emerge as critical determinants. The judicious calibration of these parameters is imperative, as it significantly shapes the models' aptitude in capturing temporal intricacies.

The management of key challenges in deploying ARIMA and LSTM models necessitates a judicious acknowledgment of their respective strengths and limitations. The study recognizes the pre-eminence of the ARIMA model in short-term forecasting, while the LSTM model exhibits promise in medium-term predictions. The comparative analysis underscores the imperative of aligning model capabilities with the temporal attributes inherent in the data.

Implications for economics and financial forecasting are profound. The insights derived from this study furnish a bespoke framework for inflation prediction, thereby offering invaluable assistance to policymakers and decision-makers. The exploration of multivariate LSTM models, enriched with additional economic indicators, presents a promising avenue for augmented precision. Furthermore, the contemplation of alternative models such as Prophet or exponential smoothing, adept at accommodating seasonality and external factors, holds potential to refine inflation projections. Such refinements, in turn, furnish indispensable tools for enlightened policymaking and decision-making within the dynamic landscape of economic considerations.

## VII. CONCLUSION

In summary, this research study has analyzed inflation prediction by implementing ARIMA and LSTM models, and the results reveals the ARIMA model's efficiency in short-term forecasting, while the LSTM model may excel in medium-term predictions. The ARIMA model exhibited impressive accuracy in forecasting short-term inflation rates, achieving an MSE of 0.0002 and an MAE of 0.012 on a holdout dataset from 2021. Conversely, the LSTM model, designed for capturing long-term dependencies, presented slightly higher predictive errors with an MSE of 0.0005 and an MAE of 0.015, suggesting its potential for medium-term forecasting. The findings emphasize the importance of aligning model choice with data characteristics, with the ARIMA model performing better on this specific dataset, underlining its suitability for short-term inflation rate prediction. These results have broader implications for economists, policymakers, and analysts, illuminating the need for careful model selection and considerations of temporal resolution in inflation forecasting tasks.

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